# Avijit Saha Asru

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## SUMMARY

PhD student in Industrial Engineering with research experience in Bayesian optimization, neural architecture search, and sequential learning. Skilled in programming with multiple technical certifications and proven leadership in academic and nonprofit settings. Currently focusing on multi-objective optimization, data-driven decision-making, and machine learning to advance innovative and impactful research.

## EDUCATION

**PhD in Industrial Engineering** Expected: July 2029
University of South Florida, Tampa, Florida, USA.

**MS in Industrial and Production Engineering** Awarded: August 2026
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. **CGPA-3.50/4.00**

**B.Sc. in Industrial and Production Engineering** Awarded: May 2023
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. **CGPA-3.56/4.00**

## RESEARCH INTEREST

- **Operations Research**
- **Bayesian Optimization**
- **Data Science and Machine Learning**
- **Multi-objective Optimization**
- **Reinforcement Learning**
- **Neural Architecture Search**

## RESEARCH EXPERIENCE

**Pessimism-Oriented Surrogate-Assisted Evolutionary Search for Bi-Objective Neural Architecture Search** 2025 - Present
**Advisor: Dr. Hadi Charkhgard, University of South Florida**

- Developed PO-NSGA, an uncertainty-aware, surrogate-assisted evolutionary algorithm for multi-objective Neural Architecture Search (NAS) and AutoML, which uses deep-ensemble predictions and pessimistic acquisition to find accurate, low-FLOPs networks for Edge AI while training only 10 of 600 candidates per generation.
- Cut search cost on CIFAR-10 by reaching NSGA-Net's Pareto front with 5.6× fewer trainings (350 vs. 1970), beating it on all six front-quality metrics (hypervolume, IGD+, etc.), and finding networks as small as 22M FLOPs. At an equal budget, its front dominated 91% of a multi-objective Bayesian optimization (qNEHVI).
- Diagnosed surrogate failure in black-box optimization as the optimizer's curse: predictors were unbiased on random networks but over-predicted selected ones by +8 points. Ran large-scale experiments in PyTorch on a distributed multi-GPU HPC cluster (SLURM); to be presented at INFORMS Annual Meeting 2026.

**A Reinforcement Learning-based Multi-step Bayesian Optimization Framework for Modeling Multi-Objective Experimentation and Sequential Decision-Making (Under Review)** 2024 - 2026

- A novel attention-based reinforcement learning (PPO) multi-step multi-objective Bayesian optimization (MOBO) framework that reformulates conventional myopic one-step acquisition as a sequential decision-making problem.
- Outperformed the state-of-the-art one-step baseline across all six optimization problems, including two synthetic benchmarks and four real MAX-phase materials-discovery tasks, achieving improved Pareto-front coverage with lower IGD.
- Improved worst-case reliability for conflicting max–min objectives by increasing failure-case hypervolume performance from approximately 77–78% to above 90–95% PHV, while reaching target Pareto-front coverage with fewer evaluations on the multimodal ZDT4 benchmark.

**Applications of Artificial Intelligence in Inventory Management** 2023 - 2024

- Conducted a systematic literature review of Artificial Intelligence (AI) and Machine Learning (ML) methods for inventory control and supply chain optimization, covering demand forecasting, deep learning, reinforcement learning, and metaheuristics.
- Performed a comparative analysis of inventory optimization models published 2015–2023, ranging from classical Operations Research models (EOQ, stochastic and multi-echelon inventory) to data-driven AI approaches, to identify research gaps and trends in data-driven decision-making.

**From Automation to Autonomy in Smart Manufacturing: A Bayesian Optimization Framework for Modeling Multi-Objective Experimentation and Sequential Decision Making** 2022 - 2024

Asru, A.S., Khosravi, H., Ahmed, I. *et al. International Journal of Advanced Manufacturing Technology*, vol. 137, pp. 5027–5057, 2025. DOI: 10.1007/s00170-025-15407-z.

- A novel multi-objective Bayesian optimization sequential learning (MOBOSL) model for progressively selecting production parameters with achieving 87% area of the design space within 25% less iteration.
- A novel parameter optimization process that can quickly and precisely solve a black box optimization problem in materials discovery problem, reducing data requirements by nearly 30%.
- A comparative mathematical analysis between Design of Experiment and sequential learning proving the worth of sequential learning by achieving 90% of PHV value in max-max and max-min optimization.

**Super Handy Lawn Mower** 2021 - 2022

- Designed a versatile lawn mower that can be operated using a cycle as a base.
- Implemented a manual paddling system as well as an electric motor for operation.
- Designed and manufactured a mechanism that allows the mower to be fully disassembled from the cycle and used as a push lawn mower.

## PROFESSIONAL EXPERIENCE

**University of South Florida** Aug 2025 - Present

***Graduate Teaching Assistant - Industrial and Management Systems Engineering*** ***Tampa, FL, USA***

- Responsibilities: Exam proctoring, Assignments and exams grading, Conducting classes
- Course Taught: Economical Analysis of Systems

**BGMEA University of Fashion and Techonology** Oct 2023 - July 2025

***Lecturer - Department of Industrial Engineering*** ***Dhaka, Bangladesh***

- Research Areas: Operations Research, Bayesian Optimization, and Data-driven Decision Making.
- Courses Taught: Operations Research, Elements of Mechanical Engineering, Engineering Materials, Computer Integrated Manufacturing, Quality Management, Measurement Instrumentation and Control.

**Rahimafrooz Batteries Limited** Apr 2022 - June 2022

***Engineering Intern - Planning and Maintenance Department*** ***Gazipur, Bangladesh***

- Redesigned the layout of a crucial manufacturing plant section with the assistance of the Relationship Diagramming and reduced 21.5% distance traveled per day.
- Conducted a root cause analysis to identify the factors contributing to the bottleneck in the assembly section, resulting in a 9.5% reduction of cycle time.
- Implemented "Safety Data Sheet" reflecting a projected 7% enhancement of safety measures within the plant.

## TECHNICAL SKILLS

- **Programming Languages:** C, Python, MATLAB
- **Simulation Software:** Arena, Ansys
- **Computer-Aided Design (CAD) Software:** SolidWorks, AutoCAD, Catia
- **Typesetting:** MS Word, LaTex
- **Other Software:** Adobe Premiere Pro, Adobe Photoshop

## CERTIFICATIONS

- Safety Management - Department of Industrial and Production Engineering, BUET — June 2022
- Python Project: pillow, tesseract and opencv - University of Michigan (Coursera Online) — Oct 2020
- Data Collection and Processing with Python - University of Michigan (Coursera Online) — Aug 2020
- Supply Chain Logistics, Planning and Operations - Rutgers Business School (Coursera Online) — July 2020
- Python Functions, Files and Dictionaries - University of Michigan (Coursera Online) — June 2020
- Student Leadership Development Workshop - U.S. Embassy, Dhaka — July 2019

## SCHOLARSHIPS/AWARDS/ACHIEVEMENTS

- Graduate Student Fellowship for academic and research excellence — University of South Florida, 2025-26
- Dean's List Award for attaining at least 3.75 GPA out of 4.00 (Level 4) — BUET, Bangladesh, June 2023
- University Merit Scholarship for being within top 10 students (Level - 3 and 4) — BUET, Bangladesh, 2021-23

- Emerging Top 10 Student Leaders — BUET, Bangladesh, 2021
- Finalist in National Business Competition — BUET Entrepreneurship Development Club, 2019

## LEADERSHIP EXPERIENCE

**INFORMS Student Chapter - University of South Florida** 2026 - 2027
***Outreach Officer*** *USF, Tampa, FL*

- Lead outreach and partnerships for the chapter, connecting students with industry professionals, alumni, and other INFORMS chapters for seminars, networking events, and career development.
- Promote Operations Research, Analytics, and AI/ML research activities on campus and across the INFORMS community, increasing student engagement and membership.

**IEOM Society International - BUFT Student Chapter** Jan 2025 - Jan 2026
***Faculty Advisor*** *BUFT, Bangladesh*

- Guided members in organizing industrial engineering–related workshops, competitions, and research activities.
- Advised the executive committee on academic and professional development initiatives, fostering collaboration between students, faculty, and industry professionals.

**IEOM Society International - BUET Student Chapter** June 2022 - June 2023
***President*** *BUET, Bangladesh*

- Served as the organization's spokesperson and established organizational policies.
- Executed 10 research workshops and 1 intra-idea competition to encourage students into research.

**BUET Film Society** Mar 2022 - May 2023
***President*** *BUET, Bangladesh*

- Established a strategic plan for renovating the club after COVID-19, resulting in a 30% revenue increase.
- Created impact programs towards the organization's vision; led the core team from 4 different creative sectors.

**Shopno Sarothi - Non Profit Organization** Nov 2018 - May 2023
***Co-founder and President*** *Dhaka, Bangladesh*

- Provided visionary leadership and developed the organization's mission, vision, and strategies. Built and led a dedicated team of over 200 volunteers all over the country, fostering a positive and inclusive culture.
- Secured about 3M BDT through strategic fundraising over 4 years and managed the organization's finances.
- Managed 1 month of food for 1000 families during COVID-19 by strategic partnerships with 5 organizations.

**Aloron - NGO** Mar 2018 - Feb 2022
***Co-founder and Head of Operations*** *Dhaka, Bangladesh*

- Organized workshops to raise awareness and provided education on cyber security to 30 adolescents.
- Spearheaded the "Freedom of Consciousness in Education" project, aiming to educate 50 working-class people.
- Deployed an emergency on-field team with 25 members, supplying essential medical equipment to 5 national hospitals and other front-line workers during the COVID-19 pandemic.

## REFERENCES

**Dr. Hadi Gard**
Associate Professor, Department of Industrial & Management Systems Engineering, USF, Tampa, Florida, USA
Email: hcharkhgard@usf.edu

**Dr. Tapas K. Das**
Professor and Head, Department of Industrial & Management Systems Engineering, USF, Tampa, Florida, USA
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